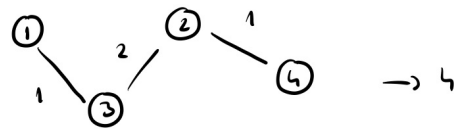
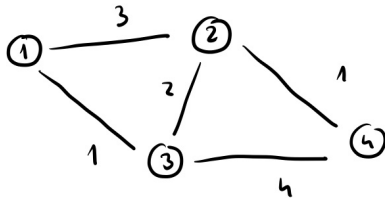


31 Oct 2025

AF Seminar 3

1. Arbori parțiali de cost minim
2. Algoritmul lui Kruskal
3. Find și Union
4. Algoritmul lui Prim
5. Exerciții APM examen + colwin

Arbori parțiali de cost minim (APM)



Algoritmul lui Kruskal

- sortăm muchiile

$$[1, 3] = 1$$

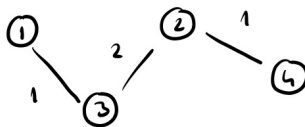
$$[2, 4] = 1$$

$$[2, 3] = 2$$

$$[1, 2] = 3$$

$$[3, 4] = 4$$

- adăugăm muchiile și verificăm **nă nu fie ciclu**



Obs : nu minimizăm distanța dintre 2 noduri
(aka drum de cost minim)

Find & Union

1) Find

2) Union

Exemplu

$n = 7$

Union 2 3

①

②

③

④

⑤

⑥

⑦

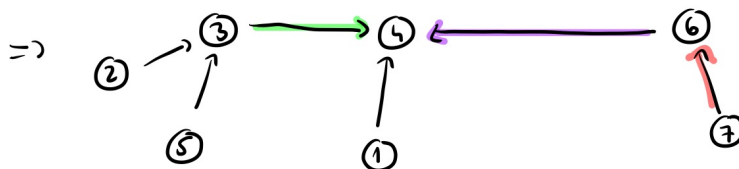
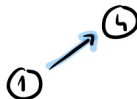
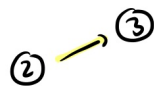
Union 1 4

Union 1 2

Union 3 5

Union 6 7

Union 1 7



Unirea după rang => $O(\log n)$

Compresie de drumuri => $O(\log^* n)$

$\approx O(1)$

Algoritmul Kruskal

```
int Find ( int x) // găsește reprezentantul
{
    while ( x != p[x] )
        x = p[x];
    return x;
}
```

$O(m \log m + m \log n)$

sortarea
măsurilor

```
int Union ( int x, int y )
{
    int a = Find ( x );
    int b = Find ( y );
    if ( a == b )
        return 0;

    else
    {
        if ( h[a] > h[b] )
            p[b] = a;
        else if ( h[a] < h[b] )
            p[a] = b;
        else {
            p[a] = b;
            h[b] ++;
        }
        return 1;
    }
}
```

```
int FindCompressie ( int x )
```

```
{
```

```
    int r = x
```

$O(m \log m + \underbrace{\log^+ n}_{O(1)})$

```
    while ( r != p[r] )
```

$= O(m \log m)$

```
        r = p[r];
```

```
    while ( x != r )
```

```
        int y = p[x]
```

```
        p[x] = r
```

```
        x = y
```

```
    return r;
```

```
}
```

```
int main ()
```

```
{
```

```
    cin >> n >> m
```

```
    for ( int i = 1; i <= n; i++ )
```

```
    {
```

```
        p[i] = i;    // parintele
```

```
        h[i] = i;    // mullimea in arbore
```

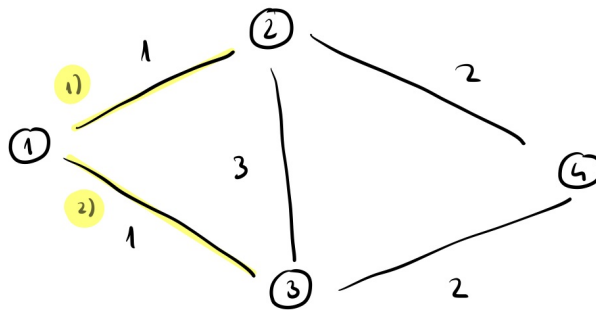
```
    }
```

```
    DSU
```

```
}
```

Algoritmul lui Prim

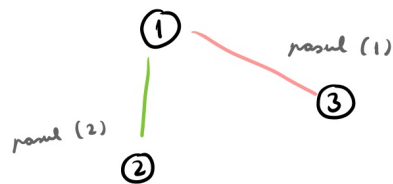
- alegem muchia de cost minim care se leagă de APM (nu neapărat de ultimul nod adăugat, fără să formeze ciclul)



Prim $O(n^2)$

$d[i]$ = muchia de cost minim legată de i

	1	2	3	4
d		1	1	2



Prim ($n \log n$)

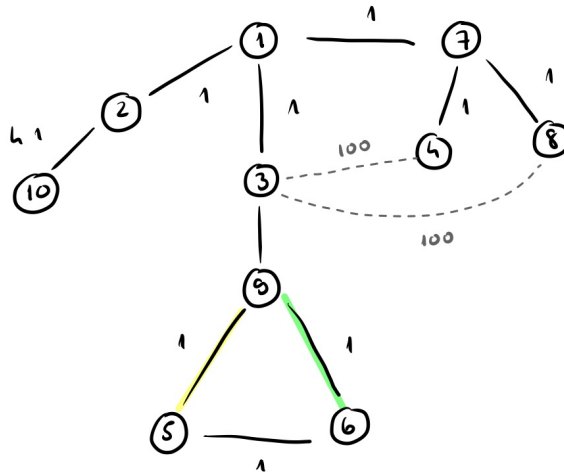
sistem minimul folosind Heap Sort

Exeritiu examen

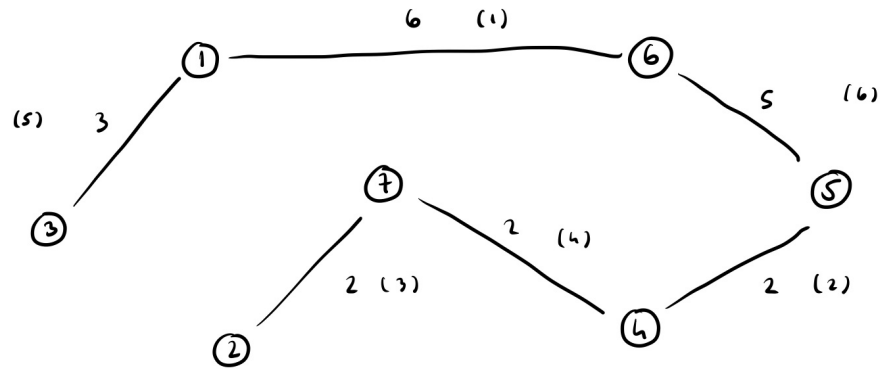
Model Examen 2022-2023 Subiect 04

ex 6

4)



6)



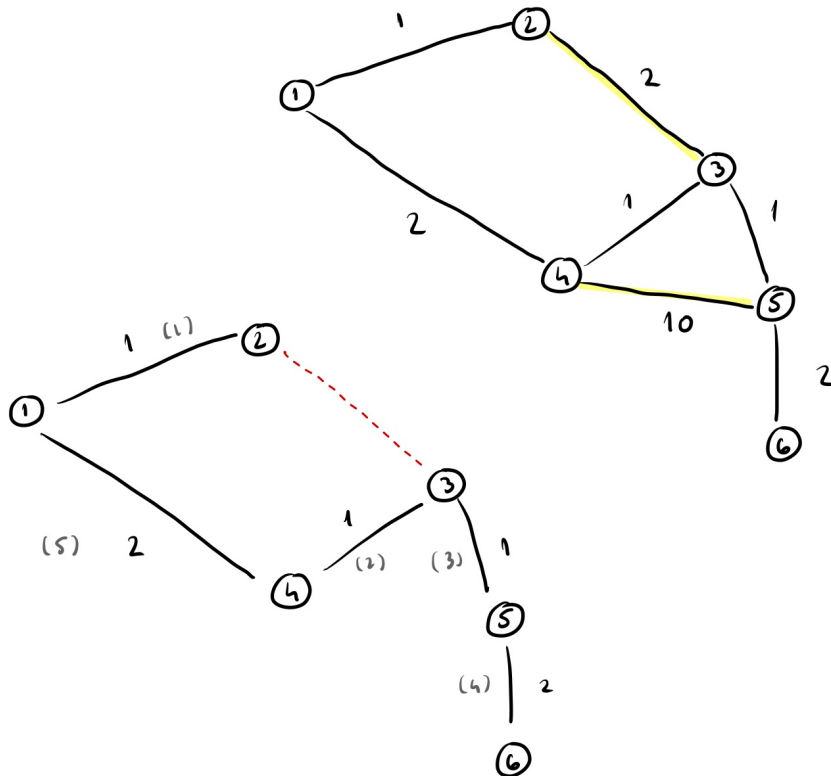
2, 4 nu adaugam $\Rightarrow$ ciclu

Conditia ca APM-ul sa fie unic $\Rightarrow$
sa nu existe 2 muchii in același cost

Exercițiu Kluwin

ex

dacă o muchie aparține unui APM
#



- construim APM

- verificăm pt fiecare muchie $\notin$ APM n_i
comparam cu muchia de cost maxim în ciclul
pe care îl include $\Rightarrow$ egal cu muchia maximă $\Rightarrow$ DA

Algoritm

$c[i][j]$ = mărimea de cost maxim de la
nodul i la nodul j

BFS / DFS pentru fiecare nod

DFS (2)

$$c[2][2] = 0 \quad \text{costul minim}$$

$$c[2][1] = \max(c[2][2], w[2,1]) = 1$$

$$c[2,4] = \max(c[2][1], w[2,4]) = 2$$

$$c[2,3] = 2$$

1						
2	1	0		2		
3						
4						
5						
6						